# A Comparative Analysis of Two Capacitor-Less LDO Topologies in 130nm CMOS Technology

Kourosh N. Baygi, *Student Member*, *IEEE*, *National Technical University of Athens*, *Greece*, *School of Electrical & Computer Engineering*, el22174@mail.ntua.gr

***Abstract*—This paper presents the design and comparison of two capacitor-less (CL) PMOS low dropout (LDO) regulators implemented in a 130nm CMOS technology. Each LDO topology was designed using a different OTA (5T OTA, Telescopic OTA). Both LDO configurations drive a 50 pF load, deliver a load current of 500 µA, support programmable output voltages (2.3 V, 2.1 V and 1.6 V), and have a consumption strictly below 100 µA. Simulation results demonstrate that the 5T OTA provides the most balanced final results, whereas the Telescopic OTA maximizes loop gain at the sever expense of the Error Amplifier's output voltage swing.**

## I. Introduction and Theoretical Framework

Low dropout (LDO) voltage regulators are one of the most essential building blocks in power-management systems. Specially with today's increasing demand for system-on-chip solutions, many mixed-signal systems utilize LDO regulators to deliver local supply voltages with reduced noise.

This work will focus on designing two individual LDO architectures. Both LDO topologies must meet the proposed specifications that are presented in Table I. The general LDO schematic is also shown in Figure 1.

TABLE I
Technical LDO Target Specifications

| Parameter | Target Value |
|---|---|
| Power Supply Voltage ($V_{EXT}$) | 3 V |
| Bias Current ($I_{bias}$) | 5 µA |
| Load Current ($I_L$) | 500 µA |
| Load Capacitance ($C_L$) | 50 pF |
| Reference Voltage ($V_{REF}$) | 1.2 V ± 1 % |
| Output Voltage ($V_{OUT}$) | 2.3 V |
| Programmable Trim Voltages | 2.1 V, 1.6 V |
| Compensation Capacitor ($C_c$) | < 7 pF |
| Phase Margin @ No-Load | > 60° |
| Phase Margin @ Full-Load | > 70° |
| PSRR @ 1 Hz | > 65 dB |
| PSRR @ 1 MHz | > 2 dB |
| PSRR @ 10 MHz | > 2 dB |
| Output-Referred Noise @ 10 Hz | $< -115\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output-Referred Noise @ 1 kHz | $< -125\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output-Referred Noise @ 10 kHz | $< -134\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output-Referred Noise @ 100 kHz | $< -143\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |
| Output-Referred Noise @ 1 MHz | $< -155\,\mathrm{dBV}/\sqrt{\mathrm{Hz}}$ |

As shown in Figure 1, a PMOS LDO is made of an Error Amplifier (EA), a PMOS Pass Device (PD), two feedback resistors $R_1$ and $R_2$ for setting the output voltage and a compensation network consisting of a nulling resistor $R_c$ and a capacitor $C_c$. In this work only one-stage EAs will be presented and analyzed.

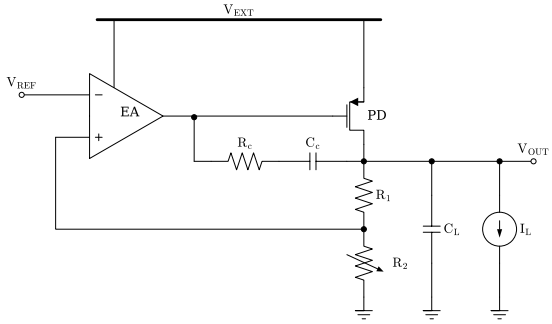



Fig. 1. General LDO Schematic

### A. Output Voltage

Assuming that the EA has infinite DC gain (therefore $A_{EA,0} \to \infty$), LDO's gain is given by:

$$\frac{V_{OUT}}{V_{REF}} = \frac{A_{EA,0}}{1 + A_{EA,0}\beta} \xrightarrow{A_{EA,0}\to\infty} \frac{V_{OUT}}{V_{REF}} = \frac{1}{\beta} \tag{1}$$

where $\beta$ is the feedback factor set by $1 + \frac{R_1}{R_2}$. Therefore:

$$\frac{V_{OUT}}{V_{REF}} = \frac{R_1}{R_1 + R_2} \to \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1 \tag{2}$$

### B. Stability

A small signal model of the LDO schematic can be seen in Figure 2 where the EA's transconductance, output resistance and output load are represented by $g_{m,EA}$, $R_1$ and $C_1$ respectively. Similarly, $g_{m,p}$, $R_2$ and $C_2$ represent the same values for the PD. The LDO's transfer function has two poles (given that we have a one stage EA), $\omega_{p1}$ (dominant) and $\omega_{p2}$ (non-dominant) and a zero $\omega_z$, all of which can be approximated with:

$$\omega_{p1} = \frac{1}{g_{m,p} R_1 R_2 C_c} \tag{3}$$

$$\omega_{p2} = \frac{g_{m,p}}{C_1 + C_2} \tag{4}$$

$$\omega_z = \frac{-1}{C_c(1/g_{m,p} - R_c)} \tag{5}$$

Even though unity gain bandwidth is not a required specification, one could find the unity gain frequency of the LDO using:

$$UGF \cong \frac{\beta g_{m,EA}}{C_2} \tag{6}$$

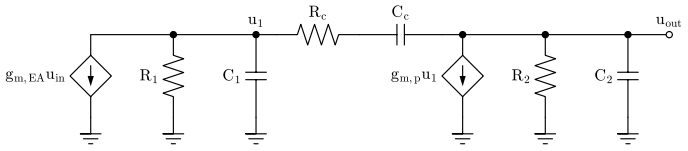



Fig. 2. Small Signal Model of LDO

### C. Power Supply Rejection Ratio (PSRR)

Power supply rejection refers to the amount of voltage ripple at the output of the LDO coming from the input voltage and therefore is of great importance. The PSR transfer function of a LDO regulator strongly depends on the type of the EA and whether the input transistors of the EA are NMOS or PMOS (Type-A or Type-B EA respectively). For a PMOS PD, Type-A EAs yield the best PSRR results at low frequencies [2]. Therefore, both LDOs presented in this work utilize Type-A EAs and their PSRR can be approximated by:

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,p} r_{ds,p}} \tag{7}$$

At higher frequencies, the compensation resistor along with the sizing of the PD can help in achieving the target PSRR requirements.

### D. Noise

Noise in LDO regulators refer to the flicker and thermal noise generated by transistors and resistors. Aside from that, the reference voltage can also contribute to the total noise in LDO. If $S_{n,ref}(f)$, $S_{n,EA}(f)$, $S_{n,PD}(f)$, $S_{n,R1}(f)$ and $S_{n,R2}(f)$ represent the noise power spectral denisty of the voltage reference, error amplifier, pass device and feedback resistors respectively, then the total output noise power spectral density of the LDO regulator is [2]:

$$\begin{aligned} S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f)) \left(1 + \frac{R_1}{R_2}\right)^2 \\ + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f) \end{aligned} \tag{8}$$

The EA and feedback resistors noise are typically the dominant sources of a LDO regulator noise.

## II. 5T OTA-Based LDO

The 5-transistor OTA used as EA in this LDO variation can be seen in Figure 3. 5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{ds,2} + g_{ds,4}} \tag{9}$$

whereas the output impedance:

$$g_{5T,out} = g_{ds,2} + g_{ds,4} \tag{10}$$

In order to satisfy both the PSRR and output voltages, the EA has to deliver enough gain. This can be achieved by increasing the transconductance of the NMOS input pair and the output resistance. Large transistors will also reduce the Flicker noise (as we see later) leading to reduced output referred noise at lower frequencies. Table II shows the device sizing for this LDO variation.

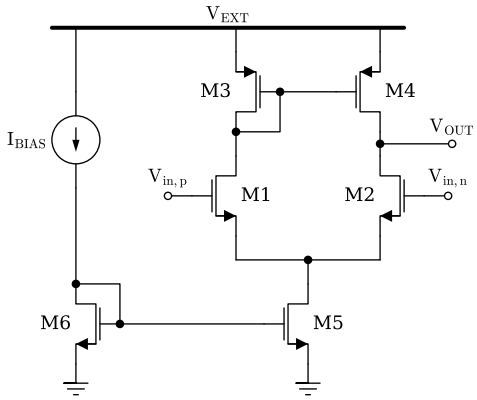



Fig. 3. 5T OTA

TABLE II
DEVICE SIZING FOR THE 5T OTA LDO

| Device | Value / Sizing | Unit | |
|---|---|---|---|
| Error Amplifier | $M_1, M_2$ (Input Pair) | 42.0/5.0 | µm |
| | $M_3, M_4$ (Active Load) | 3.8/5.0 | µm |
| | $M_5$ (Tail Current) | 12.0/5.0 | µm |
| Power Stage | $M_{PD}$ (Pass Device) | 20.0/0.14 | µm |
| Passive Network | $R_1$ (Divider) | 15.00 | kΩ |
| | $R_{21}$ (Divider) | 16.36 | kΩ |
| | $R_{22}$ (Divider) | 20.00 | kΩ |
| | $R_{23}$ (Divider) | 45.00 | kΩ |
| | $R_c$ (Compensation) | 15.00 | kΩ |
| | $C_c$ (Compensation) | 7.00 | pF |

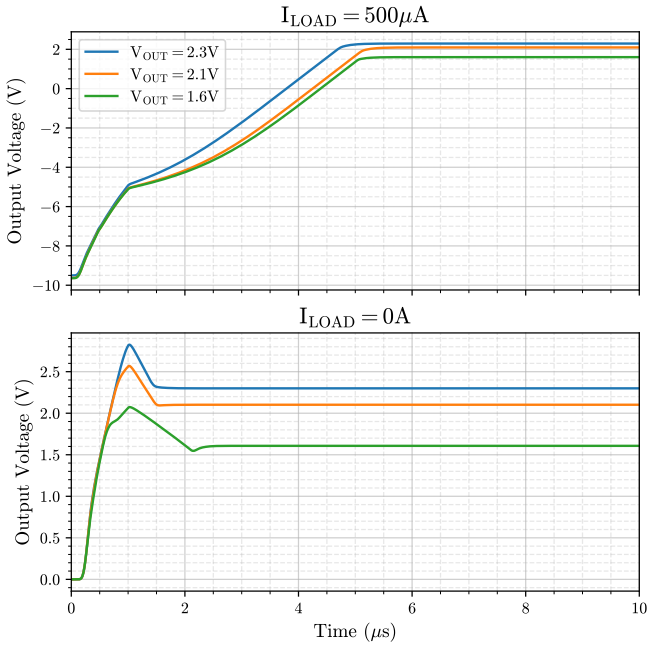



Fig. 4. 5T OTA-Based LDO's Ramp Response

The first thing to see is whether this topology can achieve the required output voltages. Figure 4 illustrates the LDO's ramp

response where we can see that the system settles at the target voltages of 2.3 V, 2.1 V, and 1.6 V with a minimal steady-state offset. This means that the EA (5T-OTA) has an acceptable DC loop gain. Both the loop gain and the loop phase are presented in Figure 6 which also shows the system's phase margin under load. By changing the value of the nulling resistor and therefore the system's zero, the phase margin can either be decreased or increased.

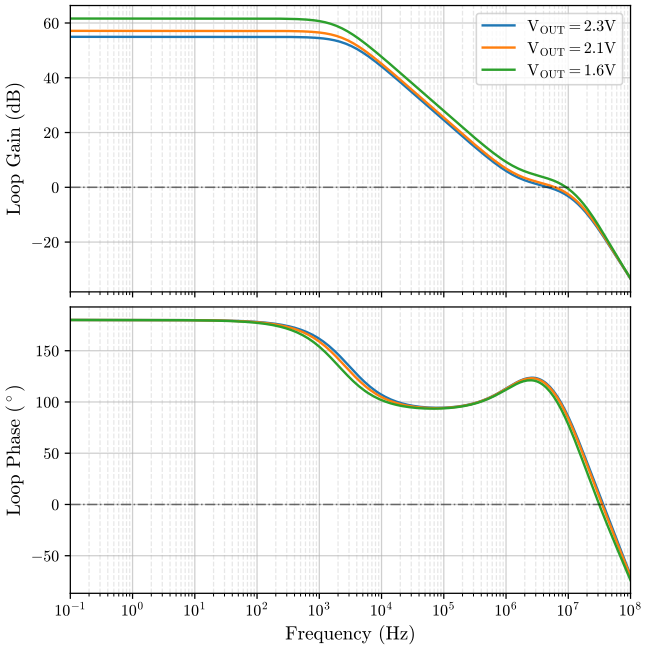



Fig. 5. 5T OTA-Based LDO's Bode Plot Under Load

Figure 6 shows the PSRR of the proposed LDO. Simulations prove that the width of the PMOS active load can directly affect the PSRR of LDO and as a result, a small PMOS active load was chosen. The nulling resistor and the sizing of the PD also help in achieving the PSRR at higher frequencies.

As for the output referred noise, as it was mentioned before, larger input transistors help in reducing the DC noise. For noise at higher frequencies, one must increase the current of the transistors, leading to a reduced transistor Thermal noise.

Finally, the simulation results for this LDO topology are presented in Table III. Note that we were unable to satisfy the noise specification at 1 MHz.

TABLE III
SIMULATED PERFORMANCE SUMMARY (ACROSS ALL TRIMS)

| **Parameter** | **Simulated Value (Min – Max)** |
|---|---|
| Trim Voltage 1 | 2.293 V – 2.299 V |
| Trim Voltage 2 | 2.096 V – 2.102 V |
| Trim Voltage 3 | 1.602 V – 1.607 V |
| Quiescent Current | 48.16 µA – 94.49 µA |
| Phase Margin ($I_{load} = 0$) | 77.62° – 99.2° |
| Phase Margin ($I_{load} = 500\,\mu A$) | 82.7° – 116.2° |

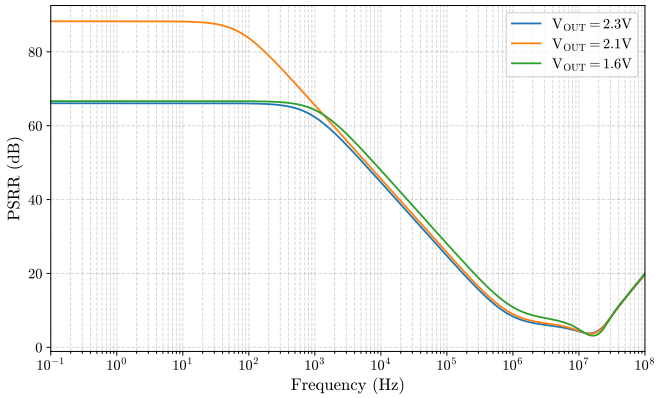



Fig. 6. 5T OTA-Based LDO's PSRR

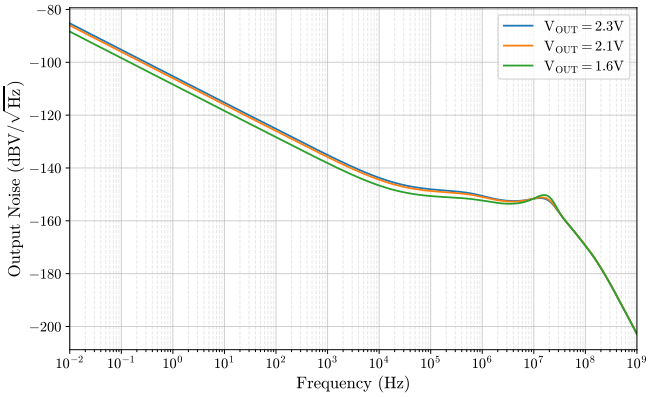



Fig. 7. 5T OTA-Based LDO's Output Referred Noise

## ACKNOWLEDGMENTS

This work wouldn't have been possible without the help of PhD candidate Andreas Papathanasiou and Pr. Paul-Peter Sotiriadis. Special thanks go to Dr. Dimitrios Baxevanakis for his guidance and help all throughout the project.

## REFERENCES

[1] T. Chan Carusone, D. A. Johns, and K. W. Martin, *Analog Integrated Circuit Design*, 2nd ed. John Wiley & Sons, Inc., 2012.
[2] J. Torres, M. El-Nozahi, A. Amer, S. Gopalraju, R. Abdullah, K. Entesari, and E. Sanchez-Sinencio, "Low Drop-Out Voltage Regulators: Capacitor-less Architecture Comparison," *IEEE Circuits and Systems Magazine*, vol. 14, no. 2, pp. 6–26, 2014.
[3] H. Pretl, M. Koefinger, and S. Dorrer. Analog (Integrated) Circuit Design. [Online]. Available: https://iic-jku.github.io/analog-circuit-design

**Kourosh N. Baygi** was born in Mashhad, Iran, in 2003 and is currently an undergraduate student in the School of Electrical & Computer Engineering in National Technical University of Athens.